



Fig. 13: The dashboard of the SDRUno GUI for the RSP1A software-defined radio

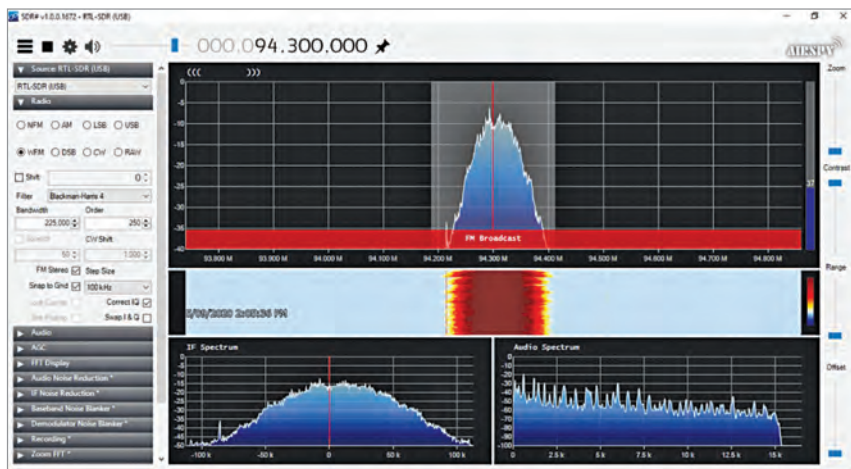


Fig. 14: SDR# software dashboard for the RTL-SDR V3 USB dongle

At one end of each of these sections, drill two 3mm diameter holes as shown in Fig. 9. Leave the 60cm length of these two sections as it is at this stage. We shall adjust the length to optimum value at a later point of time.

It is strongly recommended to use a general-purpose PCB or any other plastic insulating material or box to mount these dipole elements for better mechanical stability and protection from the weather and dust/moisture. The exact dimensions and construction details are shown in Fig. 9. The author terminated the connections to the dipole elements into a BNC header suitably mounted on the PCB but a panel mount type would also serve the purpose.

Cable connections. Since this

linear dipole offers impedance of about 72 ohms, the commonly available 75-ohm co-ax cable used for satellite television can be used here. You may need about 12 metres of length. You can choose as per your needs but do not exceed 15 metres to avoid significant signal losses due to the cable. Terminate one end of this cable into BNC male plug and the other end into a BNC cable-type socket. Refer Fig. 11 for details. These are easily available in local electronics/CCTV shops. Also, as can be seen in Fig. 9 and Fig. 10, the coax cable is coiled into 3-4 turns near the antenna. This acts as a balun to take care of impedance transformation/matching.

Antenna mount. After your antenna (dipole) assembly is complete,

it is time to mount it on the mast. Theoretically and ideally, we need the antenna mounted as high as possible. But there are two limiting factors. First, as stated above, is the cable loss. And the second is the cost of material for the mast and eventually the mechanical stability against winds. The author used just a random length of about 210cm that was available as two surplus 1.27cm dia PVC pipes left over after household plumbing work. See Fig. 9 and Fig. 10 for an idea about the mast. The author had the benefit of a rooftop two stories high.

You are at liberty to choose your own material for the mast as long as it is insulating and stays within your budget. One final word—place this antenna away from power lines and other sources of electrical or RF interference. Any kind of electrical noise or interference affects the image quality—sometimes no data at all!

Low noise amplifier (LNA). This is the antenna front-end amplifier that raises the signal level by 10 to 20dB. At the beginning stage you need not use an LNA, but it will be useful for receiving the high-quality images from the Russian Meteor M2 satellite. The satellite transmits weather images in the same 137MHz band at 137.100MHz (as received by the author) but not in the analogue APT format. It uses QPSK modulation, with symbol/data rate is at 72kbps, and requires different demodulation approach.

So, for the moment, the LNA is optional. It can however be constructed on your own or bought off-the-shelf. The author built one such LNA, which is under evaluation and testing. A significant cost gets involved with this as microwave/RF components are required to achieve the desired performance.

Receiver. After the antenna, the main part of the receiving station is the receiver. There are affordable wideband software-defined radio